

Curated Research Articles

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- **Granular Creep and Its Role in Optimizing Solid Electrolyte Fabrication for All-Solid-State Batteries** — score: 1.000 This study explores the impact of granular creep on the densification of Li₆PS₅Cl solid electrolyte separators used in all-solid-state batteries. Through simulations and experiments, it demonstrates that low strain rates facilitate particle rearrangement and porosity reduction, resulting in improved ionic conductivity and significantly higher critical current density compared to traditionally pressed separators. The findings emphasize the critical role of controlled strain rates in optimizing the microstructure and electrochemical performance of solid electrolytes. Journal: *Wiley: Small Methods: Table of Contents*
- **High-Resolution Mapping of Discharge Product in Li O₂ Batteries** — score: 1.000 The study investigates the discharge product distribution in Li O₂ batteries using advanced imaging techniques, identifying oxygen mass transport as the key factor limiting battery performance. By mapping discharge products within air electrodes, the research reveals that accumulation at the oxygen side leads to pore blockage and reduced capacity during high-rate discharge, thus highlighting critical areas for improving battery efficiency. Journal: *Wiley: Small Methods: Table of Contents*
- **Stabilized O₃-Type Layered Oxide Cathode via High-Entropy Engineering** — score: 0.900 The article presents a novel high-entropy O₃-type layered oxide cathode (NaNi_{0.24}Fe_{0.24}Mn_{0.24}Li_{0.07}Mg_{0.07}Ti_{0.07}Sb_{0.07}) designed to enhance the stability and performance of sodium-ion batteries. Through multi-cation synergy and entropy-driven design, this cathode exhibits significant improvements in phase transition reversibility and cycling stability, outperforming traditional benchmarks while achieving long-term retention and superior rate capability. Journal: *Wiley: Small Methods: Table of Contents*
- **Electrostatic Dry Coating Strategies for Through-Plane Control of Electronic Pathways in Lithium-Ion Electrodes** — score: 0.900 The article presents a novel electrostatic dry coating method that aligns carbon fibers in lithium-ion battery electrodes to improve electronic conduction while maintaining ionic transport. This technique significantly reduces electronic resistance and enhances battery performance, offering a sustainable approach for producing high-energy-density batteries. Journal: *Wiley: Small Methods: Table of Contents*
- **MXene-Derived LiV₃O₈ Nanorods for High-Performance Aqueous Ammonium-Ion Batteries** — score: 0.900 The study presents the development of LiV₃O₈ nanorods from MXene through thermal conversion, which serve as effective anodes for aqueous ammonium-ion batteries. These nanorods achieve high capacities and excellent cycling stability by facilitating H⁺/NH₄⁺ co-intercalation, while their dynamic structural evolution enhances ion transport and surface area, promising significant advances in energy storage performance. Journal: *Wiley: Small Methods: Table of Contents*
- **Enhanced Structure Ductility and Sodium Storage Kinetics of High-Capacity Sulfide Anodes via Medium-Entropy Engineering** — score: 0.900 The article presents a novel medium-entropy ductility engineering approach to enhance the mechanical properties and performance of metal sulfide anodes in sodium-ion batteries. This method improves lattice ductility, reduces stiffness, and facilitates rapid Na⁺ diffusion, leading to exceptional cycling stability and retention of 92% capacity after 900 cycles at high discharge rates. Journal: *Wiley: Small Methods: Table of Contents*
- **Continuous Synthesis of Carbon-Coated Na₃V₂(PO₄)₃ by Segmented Flow Tubular Reactor** — score: 0.800 The article presents a novel continuous synthesis method for the sodium-ion battery cathode material Na₃V₂(PO₄)₃ using a segmented flow tubular reactor (SFTR), which significantly streamlines the production process compared to traditional hydrothermal techniques. The study includes phase formation analysis through in situ XRD and shows that the SFTR-derived material demonstrates superior electrochemical performance with minimal capacity fading, showcasing the method's potential for scalable production of advanced battery materials. Journal: *Wiley: Small Methods: Table of Contents*

- **Unlocking Limited Na-Ion Transport Channel in Na₂Fe₂(SO₄)₃ for Ultrafast Na-Ion Transport** — score: 0.800 The study demonstrates that co-doping Na₂Fe₂(SO₄)₃ with Ca, Mn, and Cu significantly enhances sodium-ion transport by elongating specific Na O bonds, effectively widening the ion diffusion channels. This results in a remarkable increase in sodium-ion diffusion coefficients and improved rate capability and cycling stability in sodium-ion batteries, highlighting the potential of multi-cation substitution for optimizing battery materials. Journal: *Wiley: Small Methods: Table of Contents*
- **Tracking the Evolution of Iridium Nanocatalysts During Acidic Oxygen Evolution Reaction by Substrate-Stabilized Identical-Location Transmission Electron Microscopy** — score: 0.800 The article discusses a novel approach using identical-location transmission electron microscopy (IL-TEM) to observe the evolution of iridium nanocatalysts during the acidic oxygen evolution reaction (OER). By employing Pt-coated grids to minimize electrode-derived artifacts, researchers successfully visualize the structural changes of individual Ir catalysts over an extended period, revealing key insights into their behavior under harsh conditions while confirming the dominance of Ir's electrochemical response. Journal: *Wiley: Small Methods: Table of Contents*
- **Fluorinated Graphite as a Fluorine Source for Mechanochemical Synthesis of Lithium Metal Oxyfluorides** — score: 0.800 The study explores the use of fluorinated graphite (CF_x) as a novel fluorine source for the mechanochemical synthesis of lithium metal oxyfluorides, addressing limitations typical of LiF-based methods, such as incomplete fluorination and impurities. Utilizing CF_x enables effective fluorine transfer through stepwise defluorination, resulting in pure oxyfluoride products with enhanced electrochemical properties, confirmed by various analytical techniques. Journal: *Wiley: Small Methods: Table of Contents*
- **Ion-Specific Templating in Deep Eutectic Solvent Directs Carbon Architectures for High-Performance Potassium-Ion Batteries** — score: 0.800 This study reveals that the type of metal ion in deep eutectic solvents significantly influences the carbonization of biomass for potassium-ion battery anodes. Specifically, calcium ions (Ca²⁺) uniquely enhance the porous structure and nitrogen content of the resulting carbon, leading to exceptional electrochemical performance, including high capacity, rapid kinetics, and impressive cycling stability. Journal: *Wiley: Small Methods: Table of Contents*
- **Ti₃C₂T_x MXene as an Active Template for Accelerated Assembly and Structuring of Electroactive Ce-MnO₂/PEDOT Nanohybrids at Bisolvent Interface** — score: 0.800 This study explores the use of Ti₃C₂T_x MXene as a novel active template for simultaneously synthesizing Ce-doped MnO₂/PEDOT nanohybrids via an accelerated liquid/liquid interfacial polymerization process. The resulting nanohybrid shows significant electrochemical sensitivity toward metronidazole, indicating its potential for pharmaceutical and environmental sensing applications, while demonstrating MXene's capability to enhance hybrid formation and structural integrity in electroactive materials. Journal: *Wiley: Small Methods: Table of Contents*
- **Correlative Chemical Imaging to Reveal the Nature of Different Commercial Graphene Materials** — score: 0.400 The study evaluates various commercial graphene-based materials using a combination of imaging techniques, primarily ToF-SIMS and SEM, to effectively analyze their morphology and chemical composition at a high spatial resolution. Additionally, it explores the impact of chemical modifications on graphene oxide samples, highlighting the challenges and potential of correlative chemical imaging in nanomaterial characterization. Journal: *Wiley: Small Methods: Table of Contents*
- **Quantifying Spin-Dependent OER Enhancement Through Butterfly-Shaped Hysteresis** — score: 0.300 The article explores the impact of magnetic fields on the oxygen evolution reaction (OER) in catalysts with in-plane magnetic anisotropy, revealing butterfly-shaped hysteresis in overpotential shifts during magnetization cycles. The study identifies spin-dependent magnetoresistance as the key mechanism behind these shifts and introduces a new metric () that quantifies the overpotential reduction per unit field, offering a way to assess the magnetic sensitivity of various electrocatalysts regardless of their bulk properties. Journal: *Wiley: Small Methods: Table of Contents*

- **Gadolinium-Enhanced Bismuth Cathode for High-Performance CO₂-to-Formate Conversion Across Electrochemical and Bioelectrochemical Energy Systems** — score: 0.300 The study introduces a gadolinium-doped bismuth catalyst (Gd-Bi@C) encapsulated in carbon, demonstrating remarkable efficiency in CO₂-to-formate conversion with a Faradaic efficiency of 95.58% and stable performance over 170 hours. This catalyst not only enhances selectivity by promoting the *OCHO intermediate* but also shows versatility in both electrochemical and bioelectrochemical systems, bridging applications in Zn-CO₂ batteries and microbial electrolysis cells for improved carbon utilization and renewable energy storage. Journal: Wiley: Small Methods: Table of Contents*
- **Flash-Combustion Synthesis of Self-Anchored Spinel Oxides via In Situ Ni Diffusion for Robust Water Oxidation** — score: 0.200 The article presents a novel flash-combustion synthesis method to create a high-performance, self-anchored Ni-Co-Fe spinel oxide electrode for water oxidation. This process enables the outward diffusion of nickel from a foam substrate, resulting in a strong interface and a porous structure that enhances oxygen evolution reaction performance while achieving significant stability. Additionally, this approach facilitates a transition to a more efficient oxygen evolution pathway, suggesting a scalable solution for sustainable hydrogen production. Journal: Wiley: Small Methods: Table of Contents
- **Correlative Chemical Imaging to Reveal the Nature of Different Commercial Graphene Materials (Small Methods 9/2026)** — score: 0.200 The authors present a novel correlative imaging technique that successfully identifies graphene flakes and impurities in complex ink matrices. By combining time-of-flight secondary ion mass spectrometry and scanning electron microscopy, they demonstrate an effective method for nanoscale visualization, enhancing the understanding of commercial graphene materials. Journal: Wiley: Small Methods: Table of Contents
- **Self-Incandescent Heating-Driven Microstructural Consolidation of Biscrolled Cu/CNT Yarns for Reduced Frequency-Dependent AC Resistance** — score: 0.200 The article discusses the use of self-incandescent heating (SIH) to enhance the electrical and mechanical properties of biscrolled Cu/CNT yarns. This method leads to improved conductivity and reduced frequency-dependent AC resistance, resulting in a strong, flexible conductor that maintains stable performance under bending and high-frequency conditions, thereby addressing common challenges faced by conventional conductive materials. Journal: Wiley: Small Methods: Table of Contents
- **HRTEM Imaging and Mechanistic Insights Into Carbon Nanotube Nucleation and Growth on Fe Nanocatalysts in a Thermal Plasma** — score: 0.200 The article details a study on the synthesis of carbon nanotubes (CNTs) using iron nanocatalysts in thermal plasma. By employing high-resolution transmission electron microscopy (HRTEM) on debundled CNTs, the researchers observed that these tubes, particularly zigzag single-wall and double-wall types, can grow from small molten Fe nanoparticles or from graphene coverings, and they present a straightforward mechanism for CNT growth involving chemical reactions at elevated temperatures. Journal: Wiley: Small Methods: Table of Contents
- **Surface Donor–Acceptor Dipole Coupling for Enhanced Conductance and Photoresponse in SnO₂ Semiconductor** — score: 0.200 The study introduces a novel surface dipole coupling strategy using electron relay-mediated interactions with redox-active viologen units to enhance the conductivity and photoresponse of tin dioxide (SnO₂). By modifying SnO₂'s surface, the research achieves a remarkable increase in electrical conductivity and extends the material's photoresponse from ultraviolet into the near-infrared spectrum, thereby improving its applicability in gas sensing, energy storage, and photocatalysis. Journal: Wiley: Small Methods: Table of Contents
- **Quantitative Analysis of Size-Dependent Structural Disorder in Ruthenium Nanoparticles by Crystal PDF Full-Space Refinement** — score: 0.200 This study employs crystal PDF full-space refinement to analyze structural disorder in ruthenium nanoparticles, revealing that their stacking sequences vary by size and phase, particularly distinguishing between face-centered cubic (fcc) and hexagonal close-packed (hcp) structures. The research highlights a correlation between nanoparticle size and stacking faults, as well as the influence of point defects on their structural properties, which is critical for understanding their catalytic behavior and guiding future targeted design of nanostructured

catalysts. Journal: *Wiley: Small Methods: Table of Contents*

- **Large-Scale MXene Membrane Fabrication via Nonsolvent Phase Separation** — score: 0.100 The article presents a novel roll-to-roll nonsolvent-induced phase separation technique for the large-scale fabrication of Ti₃C₂Tx/polysulfone mixed-matrix ultrafiltration membranes. This method significantly enhances membrane performance by improving hydrophilicity and reducing fouling, achieving a water flux of 206 LMH and 97.6% rejection of humic acid while maintaining low variability across batches. The research outlines critical parameters for practical manufacturing, advancing the potential for MXene membranes in water purification applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Low-Dose Electron Total Scattering Analysis Resolves Non-Crystalline Phase Separation in Polymer Semiconductors and Device Multilayers** — score: 0.000 The article discusses a novel approach using low-dose electron total scattering analysis to characterize phase separation in polyfluorene-based polymer semiconductor blends, focusing on both amorphous and crystalline domains. The methods allow for detailed mapping of phase distributions in both planar and cross-sectional views of devices, revealing insights into the interrelations of phase separation and crystallization that impact the performance and longevity of organic optoelectronic devices. Journal: *Wiley: Small Methods: Table of Contents*
- **Synergetic Effects of Alternating Oxidant and Carbon Supply to Enhance Defect Healing of Single-Walled Carbon Nanotubes** — score: 0.000 The article presents a novel CO₂-assisted multiple-cycle healing strategy to effectively repair defects in single-walled carbon nanotubes (CNTs) while preserving their structural integrity. By leveraging CO₂ oxidation in conjunction with acetylene (C₂H₂), this method enhances the crystallinity of CNTs without causing undesirable tube coalescence, demonstrating high reproducibility and broad applicability across various CNT types. Journal: *Wiley: Small Methods: Table of Contents*
- **Enhancing Dose Efficiency of Optimum Bright-Field Scanning Transmission Electron Microscopy Using a Phase-Shifted Electron Probe** — score: 0.000 This study addresses the challenge of imaging beam-sensitive materials in scanning transmission electron microscopy (STEM) under low electron dose conditions, which typically suffer from low signal-to-noise ratios. By applying phase shifts through lens aberrations, the authors enhance low-frequency contrast in optimum bright-field STEM, thereby improving the visibility of structural features without introducing artifacts, facilitating better imaging of materials like zeolites. Journal: *Wiley: Small Methods: Table of Contents*
- **Surface Amination for Site-Selective Activation of Cd Centers in High-Entropy Sulfide Photocatalysts** — score: 0.000 The article discusses how surface amination of high-entropy sulfide nanosheets, particularly activating cadmium centers with amino ligands, significantly enhances their photocatalytic hydrogen evolution efficiency. The modified sulfides show a remarkable 16.25-fold increase in performance by promoting charge separation and improving surface reaction kinetics through engineered electronic structures, demonstrating a novel strategy for optimizing catalytic activity in these materials. Journal: *Wiley: Small Methods: Table of Contents*
- **TiO₂ Aerogels With 2D Holey Building Blocks** — score: 0.000 The article discusses the successful synthesis of a novel TiO₂ aerogel composed of graphene-like holey nanosheets, utilizing an innovative one-step method that combines chemical vapor infiltration and template removal. This aerogel features a unique 2D holey structure that enhances its surface area and active sites, leading to superior photocatalytic performance and UV shielding compared to traditional TiO₂ aerogels, while also providing a potential application in wave-transparency materials. The approach represents a significant advancement in the fabrication of porous nanostructures for various applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Programmable Circular Single-Stranded DNA Acts as Recyclable Anti-miRNA Nucleotides to Inhibit Colorectal Cancer** — score: 0.000 The article presents a novel therapeutic approach for colorectal cancer (CRC) using programmable circular single-stranded DNA (CssDNA) as a recyclable anti-miRNA nucleotide. This CssDNA effectively targets and degrades multiple oncomiRs, simultaneously enhancing tumor suppressor gene expression while maintaining high stability and low

immunogenicity, thus offering a potential multi-targeted strategy in CRC treatment. Journal: *Wiley: Small Methods: Table of Contents*

- **Molecularly Engineered 2D Xene Heterostructures With Implanted Peroxidase-Like Activity for Mimicking Enzyme Functions** — score: 0.000 The article discusses a novel approach to creating 2D Xene heterostructures by covalently attaching a Co(II) molecular catalyst to allyl germanane, resulting in a nanozyme with peroxidase-like activity. This engineered material demonstrates high sensitivity and selectivity for detecting hydrogen peroxide, making it suitable for monitoring H₂O₂ production during enzymatic reactions in physiological environments, thus showcasing the potential of programmable 2D nanomaterials in catalysis. Journal: *Wiley: Small Methods: Table of Contents*
- **Spatially Profiling Trace Cytokine Signatures From Microscopically Derived Skin Samples to Probe Skin Disease Inflammation** — score: 0.000 This study presents a novel digital SERS immunoassay that enables the multiplex detection of pro-inflammatory cytokines IL-17A, IL-22, IL-23, and TNF within microscopically obtained psoriasis biopsy samples, achieving unprecedented sensitivity down to 104 aM. The assay's ability to differentiate cytokine profiles between perilesional and lesional skin highlights its potential for advancing diagnostic and therapeutic practices in inflammatory skin diseases. Journal: *Wiley: Small Methods: Table of Contents*
- **Highly Aqueous Stability of Metal Oxides Encapsulated CsPbBr₃ Nanocrystals via A Generalized Core-Shell Engineering** — score: 0.000 This study introduces a core-shell engineering method to enhance the stability of metal oxide-encapsulated CsPbBr₃ nanocrystals (NCs) for photocatalytic applications. The resulting MOx@CsPbBr₃ NCs, specifically SnO₂@CsPbBr₃, exhibit significant water dispersibility and stability, yielding a notable ammonia production rate of 47 $\mu\text{mol g}^{-1} \cdot \text{h}^{-1}$. This approach not only improves the resilience of perovskite materials in aqueous environments but also opens avenues for their use in nitrogen fixation photocatalysis. Journal: *Wiley: Small Methods: Table of Contents*
- **Application of Ag-Enriched Nanostructures Through Ga⁺ Irradiation of Silver Butyrate Films in Surface-Enhanced Raman Spectroscopy (SERS)** — score: 0.000 The article presents a method for creating silver-enriched nanostructures through Ga⁺ irradiation of silver butyrate films. By optimizing various parameters, researchers achieved highly conductive films and SERS substrates, demonstrating significant signal enhancement in Raman spectroscopy. This technique offers a cost-effective solution for producing microstructures for applications in sensing and electronics. Journal: *Wiley: Small Methods: Table of Contents*
- **In Situ Emulsion Crystallization Combined With Deep Potential Molecular Dynamics: Desensitization of Energetic Perovskite DAP-4** — score: 0.000 The study introduces a novel method combining in situ emulsion crystallization and deep potential molecular dynamics to successfully desensitize the energetic perovskite DAP-4. By incorporating stearic acid within microreactors and utilizing advanced simulations, the research demonstrates a significant reduction in the mobility of reactive ions, leading to enhanced safety against mechanical processing hazards while maintaining combustion efficiency. Journal: *Wiley: Small Methods: Table of Contents*
- **Simultaneous Substrate Debonding and Device Annealing of Flexible a-IGZO Thin-Film Transistors on Colorless Polyimide via Single-Shot Xenon Flash Lamp Irradiation** — score: 0.000 The article presents a novel xenon flash lamp debonding process that facilitates the simultaneous mechanical release and photothermal annealing of amorphous indium gallium zinc oxide (a-IGZO) thin-film transistors on flexible colorless polyimide substrates. This method not only enhances electrical performance by improving charge mobility and reducing off-state current but also ensures the devices maintain excellent characteristics over extensive bending cycles, making it a promising approach for advancing flexible electronics. Journal: *Wiley: Small Methods: Table of Contents*
- **Strategic Chemical Vapor Deposition of Two-Dimensional Bismuth Oxyselenide** — score: 0.000 This study explores the growth kinetics of two-dimensional bismuth oxyselenide (Bi₂O₂Se) through optimized chemical vapor deposition (CVD) methods, focusing on factors like substrate characteristics and growth conditions. A novel two-step selenization technique reveals intricate orientation-

dependent morphologies, contributing to a better understanding of Bi₂O₂Se synthesis and its potential for future electronic and optoelectronic applications. Journal: *Wiley: Small Methods: Table of Contents*

- **Bifunctional Interface Synergy for High-Performance Wide-Bandgap and Tandem Perovskite Solar Cells** — score: 0.000 This article presents a novel passivation strategy using sulfaguandine and 1,3-diaminopropane dihydroiodide to enhance the efficiency of wide-bandgap perovskite solar cells by creating a stable interface that reduces non-radiative recombination and boosts charge transport. The resulting modifications lead to a top cell with an efficiency of 20.27% and an impressive overall tandem cell efficiency of 28.14%, while also demonstrating notable stability over time. Journal: *Wiley: Small Methods: Table of Contents*
- **Direct Measurement of Protein Corona Thickness on Nanomaterials Using Dual-Color Stochastic Optical Reconstruction Microscopy (STORM)** — score: 0.000 The study introduces a novel dual-color STORM imaging technique that allows for the direct measurement of protein corona thickness on nanoparticles within biological media, revealing a consistent thickness of approximately 50 nm across various nanoparticle types. This method enhances our understanding of the spatial structure of protein coronas and establishes a reliable approach for characterizing nanomaterial-protein interactions, supported by robust validation and proteomic analysis. Journal: *Wiley: Small Methods: Table of Contents*
- **Issue Information** — score: 0.000 The abstract appears to be missing its content, so I cannot provide a summary. If you can share the text of the abstract or provide more details, I would be happy to help summarize it. Journal: *Wiley: Small Methods: Table of Contents*
- **Advancements in Free-Standing Graphene Synthesis Using Electrodeless Microwave Plasma** — score: 0.000 This review article delves into the synthesis of free-standing graphene via electrodeless microwave plasma (EMP), presenting its advantages over traditional production methods. It discusses the operational mechanisms and critical parameters influencing graphene formation, highlights the sustainability and scalability potential of EMP, and emphasizes the need for advanced techniques to optimize production efficiency while addressing existing knowledge gaps in this field. Journal: *Wiley: Small Methods: Table of Contents*
- **Advances in Axial Resolution Strategies for Super-Resolution Imaging Systems** — score: 0.000 This review article discusses the advancements in axial super-resolution imaging technologies, essential for reconstructing the 3D internal structures of biological cells. It categorizes these techniques into four types, emphasizing their principles and developments, while also offering perspectives on future advancements in the field. Journal: *Wiley: Small Methods: Table of Contents*
- **GatorST: A Versatile Contrastive Meta-Learning Framework for Spatial Transcriptomic Data Analysis** — score: 0.000 GatorST is a novel framework for analyzing spatial transcriptomic data that integrates graph-based modeling and meta-learning to effectively capture both local and global spatial contexts. By utilizing two-hop subgraph convolution, cluster-aware pseudo-labels, and episodic training, GatorST delivers enhanced representations that improve key tasks such as spatial domain detection, gene expression imputation, and trajectory inference, outperforming fifteen existing methods in these areas. Journal: *Wiley: Small Methods: Table of Contents*
- **High-Power Performance of Textured Piezoelectric Ceramics Through Synergistic A-Site Donor and B-Site Acceptor Doping** — score: 0.000 The study presents a novel co-doping strategy for PIN-PMN-PT piezoelectric ceramics that enhances both piezoelectric response and mechanical quality by utilizing Eu and Mn ions. This approach, combined with templated grain growth, yields high-performance ceramics with exceptional properties—such as a d₃₃ of 508 pC/N and Q_m over 850—while ensuring thermal stability above 170°C, paving the way for advanced high-power applications in transducer technologies. Journal: *Wiley: Small Methods: Table of Contents*
- **Synergistic Dual Engineering of NiOx via Ethylenediamine Chelation and -Ray Irradiation for Enhanced Interfacial Transport in Perovskite Solar Cells** — score: 0.000 This article presents a dual-engineering approach to enhance nickel oxide (NiOx) as a hole transport layer in per-

ovskite solar cells through ethylenediamine chelation and γ -ray irradiation. This combined method reduces defects in NiOx, boosts charge transport, stabilizes energy alignment with perovskites, and results in solar cells achieving a power conversion efficiency of 24.06% alongside improved stability and reduced hysteresis. Journal: *Wiley: Small Methods: Table of Contents*

- **Defect Suppression Strategy Targeting Oxygen-Related States in Sb₂(S,Se)₃ Thin-Film Solar Cells** — score: 0.000 The study investigates the impact of post-annealing environments on the microstructure and defect properties of Sb₂(S,Se)₃ thin-film solar cells. Nitrogen annealing promotes crystallinity and reduces defects, leading to a significant enhancement in device performance, with a peak power conversion efficiency of 8.84%. In contrast, low-vacuum conditions are shown to foster oxidation and void formation, detrimental to charge transport. Journal: *Wiley: Small Methods: Table of Contents*
- **On the Overlooked Effect of Counter-Cations From Borohydrides for the Gold- and Metal-Mediated Catalytic Reduction of 4-Nitrophenol** — score: 0.000 The study highlights the significant impact of counter-cations from borohydrides on the gold-mediated catalytic reduction of 4-nitrophenol, demonstrating that the catalytic efficiency is enhanced in the presence of potassium ions compared to lithium and sodium ions. The findings establish that potassium borohydride yields the highest catalytic activity, stressing the importance of cation selection in optimizing this reduction process for water treatment applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Manipulating MoS₂ Thickness On-Chip via Van Der Waals Delamination** — score: 0.000 The article presents an innovative van der Waals etching technique for the on-chip manipulation of MoS₂ thickness, allowing for controlled removal of layers with atomic precision and minimal surface roughness (0.21 nm). By using various metals to facilitate the peeling process, researchers can achieve programmable thicknesses ranging from 2 to over 50 nm, significantly influencing the electronic properties of these two-dimensional materials. Journal: *Wiley: Small Methods: Table of Contents*
- **Biomass Polymer-Stabilized Hygroscopic Salts in Cellulose Foams for Durable Atmospheric Water Harvesting** — score: 0.000 The article presents a novel LiCl+CMC@cellulose composite foam developed through a scalable process that enhances atmospheric water harvesting (AWH) by improving salt stability and reducing leakage. The inclusion of carboxymethyl cellulose (CMC) as a stabilizer allows for high water uptake and retention over multiple cycles, while a photothermal layer facilitates efficient solar-driven water release, offering a durable and eco-friendly solution to combat freshwater scarcity. Journal: *Wiley: Small Methods: Table of Contents*
- **Digital SERS Nanoprobe Immunoassay With Single-Particle Sensing Resolution for Detection of Pancreatic Cancer Biomarkers From Clinical Blood Samples** — score: 0.000 This study presents a novel Ag nanogap shell (AgNGS) nanoprobe-based digital SERS immunoassay capable of quantitatively detecting pancreatic cancer biomarkers CA19-9 and APOA1 in clinical serum samples. By combining advanced single-particle sensing with machine learning analysis, the assay demonstrated an impressive AUC of 0.988 for distinguishing early-stage pancreatic cancer patients from healthy individuals, showcasing its potential as a liquid biopsy approach for disease biomarker profiling. Journal: *Wiley: Small Methods: Table of Contents*
- **On the Interaction of the Photovoltaic Response With Ferroelectric and Magnetic Domains: Magnetoelectric Control of the Photovoltaic Response in BiFeO₃ Thin Films** — score: 0.000 This research explores the tunable photovoltaic properties of pure, polycrystalline BiFeO₃ thin films, revealing that external electric and magnetic fields can significantly enhance photogeneration by up to seven times. The study highlights the interplay between ferroelectric and magnetic domain orientations, suggesting that BiFeO₃ could lead to cost-effective, multifunctional photovoltaic devices with potential applications in sustainable energy through non-volatile control systems. Journal: *Wiley: Small Methods: Table of Contents*
- **Temperature-Mediated Defect Engineering in Cu₂-xS for Tailoring Dielectric Response Properties** — score: 0.000 The study presents a temperature-mediated defect engineering approach to synthesize single-phase Cu₂-xS, revealing how varying intrinsic defect types and concentrations

influence its dielectric properties. By optimizing the synthesis temperature, the researchers achieved a significant enhancement in microwave absorption capabilities, demonstrating an outstanding reflection loss and a wide effective absorption bandwidth, thereby establishing a clear correlation between defect structure and dielectric performance in this material. Journal: *Wiley: Small Methods: Table of Contents*